User Manual
Group 8 Electronic Workbench
Outlined Usage For the Whole Workbench

Lego Kit for Newbies: What we built, How to run it, and How not to brick it



Table Of Contents

Getting Started 3

Safety Rules 4

Rotary Encoder Control Settings 5

Navigation Menu Interface 6

Function Generator 7

- Purpose
- Information
- How to Use

Ohmmeter 8

- Purpose 9
- Information
- How to Use

Voltmeter 10

- Purpose
- Information
- How to Use

DC Reference 11

- Purpose
- How to Use

Frequency Measurement 12

- Purpose
- How to Use

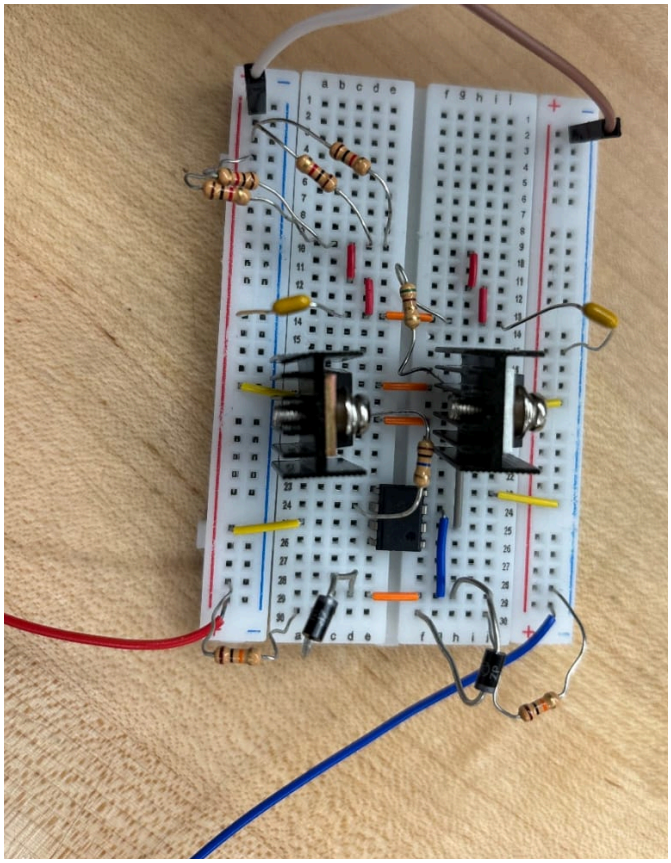
Back & Main Buttons 13

Troubleshooting / Common Bugs (FAQ) 14

Getting started

1. Connect the twin-wire barrel to the 24V power supply cable.
2. Connect the red 24V wire and blue ground wire to the breadboard as shown in the following photo.

   NOTE: FLIPPING POWER WIRES WILL DAMAGE THE DEVICE



3. Use the USB-C to power the Raspberry Pi 4 directly.
4. Wait for the response of the LCD screen, and once on with the menu, you are ready to test and play!
5. For any errors, refer to the FAQ to troubleshoot problems.

IMPORTANT — Read Before Using!

Safety Rules

These rules protect both you and the equipment. Breaking them may result in damaging the workbench or personal injury.

- Only use 3.3V logic signals on the Pi GPIO pins — NEVER apply 5V directly!
- Make sure ALL grounds are common (connected) before powering on.
- Always shut down the Pi properly before unplugging power: sudo shutdown now
- Verify multimeter calibration before resistance testing.
- Keep the back of the rotary encoder covered so skin does not short the contacts.
- Never connect live circuits while the Pi is still booted and running.
- Do not unplug power during file writes — this can corrupt the SD card and Git repo.

You're all set to start learning each of the bench's core functions.



Rotary Encoder Control settings

Turn Clockwise (slow)	Scroll DOWN the menu or increase a value by 1 step
Turn Clockwise (fast)	Jump DOWN quickly — increases by 10 steps at once
Turn Counter-Clockwise	Scroll UP the menu or decrease a value
Short Press (< 1 second)	SELECT the highlighted option or CONFIRM a value
Long Hold (1–3 seconds)	CANCEL a value entry and go back to what it was
Very Long Hold (> 3 seconds)	Go back to the MAIN MENU from anywhere + turns off all outputs

Navigation Menu Interface

a. The Main level UI is:
- i. OFF: Everything is off
- ii. Mode Select
  1. Function Generator
     - a. Type
       - i. Sine
       - ii. Square
       - iii. Back
       - iv. Main
     - b. Frequency
       - i. Input Frequency
       - ii. Back
       - iii. Main
     - c. Amplitude
       - i. Input Amplitude
       - ii. Back
       - iii. Main
     - d. Output
       - i. On (note if you back out of this or go to main, must turn off by default). UI must show value of generated Output.
       - ii. Off
       - iii. Back
       - iv. Main
  2. Ohmmeter (UI shows value of reading)
     - a. Back
     - b. Main
  3. Voltmeter (UI show value of reading)
     - a. Source
       - i. External
       - ii. Internal Reference (If selected, go to DC Reference)
       - iii. Back
       - iv. Main
  4. DC Reference
     - a. Voltage Value Input
     - b. Output
       - i. On (note if you back out of this or go to main, must turn off by default). UI must show value of generated output using your Voltmeter!)
       - ii. Off
       - iii. Back
       - iv. Main
     - c. Back
     - d. Main
  5. Frequency Measurement
     - a. Back
     - b. Main
  6. Back
  7. Main



The 7 Modes Inside Mode Select

Opens a list of 7 tools to choose from. Turn the knob to highlight one, then press to enter it.

Function Generator

Purpose

- [ ] Produce a Square or Sine wave output based on manipulated amplitude or frequency set by the user.

Information

- [ ] Can output Square or Sine wave
- [ ] Output is turned off automatically if “Back” is pressed or button is long-pressed for “return to menu” function
- [ ] Square Wave Limits
    - [ ] Frequency Range: 100 Hz - 10k Hz
    - [ ] Amplitude Range: 0-10 Vpeak
- [ ] Sine Wave Limits
    - [ ] Frequency Range: 1k Hz - 10k Hz
    - [ ] Amplitude Range: 0-10 V (positive only)

How to Use

1. Turn rotary encoder to scroll onto “Mode Select”
2. Press rotary encoder button to select “Mode Select”
3. Scroll down to “Function Generator” and press button
4. Select Waveform type (Square or Sine) by scrolling and pressing the button to select
5. Set frequency by scrolling to desired value
    a. Press button to select
    b. Hold button to cancel
6. Set amplitude to by scrolling to desired value
    a. Press button to select
7. Connect Oscilloscope probes to wires labeled “function generator”
8. Scroll to output option (if needed)
9. Press rotary encoder button to turn output on
10. View oscilloscope for generated wave output
11. To repeat:
    a. Scroll down and select “back”
    b. Repeat steps 3-10

Pictures/Video:
https://www.youtube.com/watch?v=IYhi88rVXT8&list=PL__Xac82jwCslwcRjHEkgNMK5rVNgswHd&index=4 (sine wave)

https://www.youtube.com/watch?v=jSTl8QOUb7Y&list=PL__Xac82jwCslwcRjHEkgNMK5rVNgswHd&index=3 (square wave)

Ohmmeter

Purpose

- [ ] Calculate the resistance of an unknown resistor and display the estimated value to the user

Information

- [ ] Display
  - [ ] R: Measured resistance shown in ohms below 1000 Ω, kΩ for values higher
  - [ ] Tol: Uncertainty - roughly ±73 Ω (minimum resolution of the system)
- [ ] Common Bugs
  - [ ] Nothing is showing on LCD -> Check the power status and connection
  - [ ] Wrong Number output -> Make sure resistor is connected properly
  - [ ] Number is changing -> Check wiring is secure

How to Use

1. Turn rotary encoder to scroll onto “Mode Select”
2. Press rotary encoder button to select “Mode Select”
3. Scroll down to “Ohmmeter” and press button
4. Connect resistor to banana clips labeled “ohmmeter”
5. Wait for system to automatically measure the resistor value
6. View LCD to see resistance (ohms) value calculated
7. To repeat:
   a. Scroll down and select “back”
   b. Repeat steps 3-6

Pictures/Video:
https://www.youtube.com/watch?v=jFf9qCSQPbY&list=PL__Xac82jwCslwcRjHEkgNMK5rVNgswHd&index=6

Voltmeter

Purpose

- ☐ Measure a DC voltage between -5v and +5v from an internal (DC reference) or external source

Information

- ☐ Ranges from -5v to +5v readings
- ☐ External is connected source from outside circuit
- ☐ Internal is reading from the DC reference connection

How to Use

1. Turn rotary encoder to scroll onto “Mode Select”
2. Press rotary encoder button to select “Mode Select”
3. Scroll down to “Voltmeter” and press button
4. Choose internal or external source
5. If external source is chosen
    a. Connect voltage source to banana clips labeled “voltmeter”
    b. Press the rotary encoder button to measure
    c. Wait for result
    d. View LCD screen to see reading of voltage between -5v and +5v
6. If internal source is chosen
    a. Refer to DC reference instructions for operation
7. To repeat:
    a. Scroll down and select “back”
    b. Repeat steps 3-6

Pictures/Video:
https://www.youtube.com/watch?v=bi9p2sj71CY&list=PL__Xac82jwCslwcRjHEkgNMK5rVNgswHd&index=1

DC Reference

Purpose

This outputs a DC voltage that you can choose from -5v to +5v. When enabled the screen shows a live voltmeter reading so you can compare what set versus what you are actually getting.

Range -5v to +5v and Scroll step size .625v.

How to Use

1. Go to mode select then to DC Reference(Turn and press enter)
2. Set the Voltage (Turn to set Voltage and press) , Next scroll to your desired voltage, Then short press to confirm.
3. Turn Output On (Turn to output and press on) The output is now active
4. Read the live lcd display. The screen displays your set target. This means the actual measured output side by side.

Ex: Insert an image from PI lcd display
DC OUTPT SET: +2.5000 V meas: +2.4795 V on > off

Warning:
IF your press back or main while output is ON, It will turn OFF automatically.

Pictures/Video:
https://www.youtube.com/watch?v=zcBzZjTuh4U&list=PL__Xac82jwCslwcRjHEkgNMK5rVNgswHd&index=1

Frequency Measurement

Purpose

This counts how many times per second a signal repeats (how frequency is measured). It watches the signals for exactly 1 second and reports the results

Range 1,00hz (1khz) to 10,000hz (10 kHz), IF number is greater then 10,000 h or less then 1khz it will display **“Out Of Range”**

How to Use

1. Connect the signal source (Attach the signal to GPIO4 (FREQ_COUNT_PIN)
   Note to self Insert gif
2. Go to mode select then Frequency measurement Turn and press to enter
3. Wait for measurement (The screen shows “Measuring for 1 second while it counts)
4. Read the screen (Please insert images Ex show F:4.500 kHz)
5. Press Back or golf the knob to exit (spin to back or long-hold to return to main)

Pictures/Video:
https://www.youtube.com/watch?v=dwaGOAAg0BM&list=PL__Xac82jwCslwcRjHEkgNMK5rVNgswHd&index=2

Back & Main Buttons

Purpose

- ☐ Back
  - ☐ Takes you to the previous screen (one level up). How: turn the knob “back” is highlighted, then press the knob.
- ☐ Main
  - ☐ Takes you all the way back to the first Main Menu (Off/ Mode select) How: Turn the knob until “Main” is highlighted. Then press the knob.
    Shortcut: Hold the Knob for more than 3 seconds from anywhere this will kill all active outputs

How to Use

1. Back
   a. Turn rotary encoder either left or right to navigate through options
   b. When **“Back”** is highlighted, click on the knob to take back to previous screen
2. Main
   a. Scroll down in current menu until “Main” is highlighted
   b. Press rotary encoder button to select - send user back to “Mode Select” menu

Trouble Shooting Common Bugs - FAQ

- If screen is black (if powered should have blue backlit)
  - Check power supply is connected and turned on
- If screen is on but nothing updates
  - Try restarting the program or reboot the Pi
- Knob does nothing or random inputs
  - Make sure the back of the rotary encoder is not being touched (the skin can cause false inputs)
- Knob scrolls too fast or skips values
  - Turn slower, fast spins trigger larger step changes
- Output is not working (no signal on oscilloscope)
  - Make sure output is set to ON
- Output turns off randomly
  - This happens if you press Back or hold the knob (this is intended behavior)
- Wrong frequency or amplitude
  - Check that values are within allowed range and properly set
- Ohmmeter shows wrong value
  - Make sure resistor is connected properly to banana clips
- Ohmmeter value is jumping
  - Check wiring is secure and not loose
- Voltmeter reads 0 or nothing
  - Make sure correct source (External/Internal) is selected
- Voltmeter wrong reading
  - Check connections and make sure voltage is within -5V to +5V
- DC reference not matching expected value
  - Small differences are normal, check measured value on screen
- Frequency measurement says “Out Of Range”
  - Input signal is too low (<1 kHz) or too high (>10 kHz)
- Program not starting on boot
  - Restart the Pi